Bastiaan Bergman

Geboren 1973

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Bastiaan Bergman

Applied Physics in Eindhoven, graduated 2004

Internship at Seagate – Ultra fast magnetization in FeRh

PhD at IBM in silicon valley – Racetrack memory

Western Digital – Producing 3 million hard drive heads per day

Atomwise – AI for Small Molecule Drug Discovery ⟸

Model Medicines – Developing a Covid19 drug

Basecamp Research – LLMs for Synthetic Protein Design

BenevolentAI – Interim role for company reorganisation

Pharmaceutic Index – (Founder/CTO) a partnering and in-licensing AI tool for pharma

Hard drive boom

Data capacity grew from ‘95 until ‘05 with 2x/year, from 10 MB to 10 GB per square inch – A very strong economic driver for the industry

Moving back during Covid19



October 6th, 2026

Guest Lecture – UvA Science park

Atomwise - AI for Drug Discovery

Small Molecule Drug Discovery

3D CNN for SMDD

99.9% accurate model, our customers say it is not good enough

Silicon Valley Economics

Drug Discovery

Proteins do everything in life, they're the basic machinery.

Carry oxygen, send distress signals, digest food, and copy DNA

Disease, usually means one of two things:

1. a protein machine has gone rogue (working overtime and causing inflammation or cancer)
2. or, an invader (like a virus) has brought its own harmful machinery

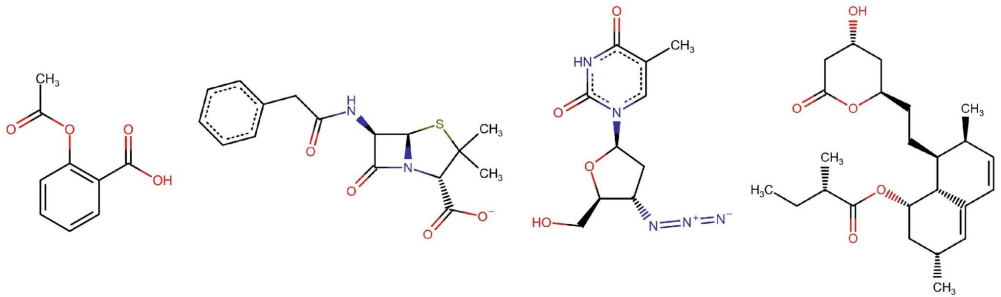
To stop them, find a molecule in one of three categories:

a. Blockers / Antagonists (The Jammed Gear)
b. Activators / Agonists (The Starter Switch)
c. Allosteric Modulators (The Monkey Wrench)

Ultimately, drug discovery is the art of finding or designing a molecule with the exact right 3D shape and electrical charge to find one specific protein out of the trillions in your body, latch onto it, and change its behavior without breaking anything else.

Small Molecules

- Molecules with a weight under 700 Daltons (Da)
- E.g. Aspirin $C_9H_8O_4$ (180 Da), Contrast: Monoclonal Antibody: 150 kDa
- Typically ‘Lipinski's Rule of 5’ – Favorable physicochemical properties such as solubility in water and fats
- How They Work (Mechanism of Action)
  - Fit precisely into a deep pocket on the target protein
  - Typically block an enzyme's function or switch a receptor on/off by binding to an active site
- Advantages:
  - Oral Bioavailability
  - Intracellular Access
  - Low Manufacturing Cost
- Limitations:
  - The "Undruggable" Challenge
  - Off target effects

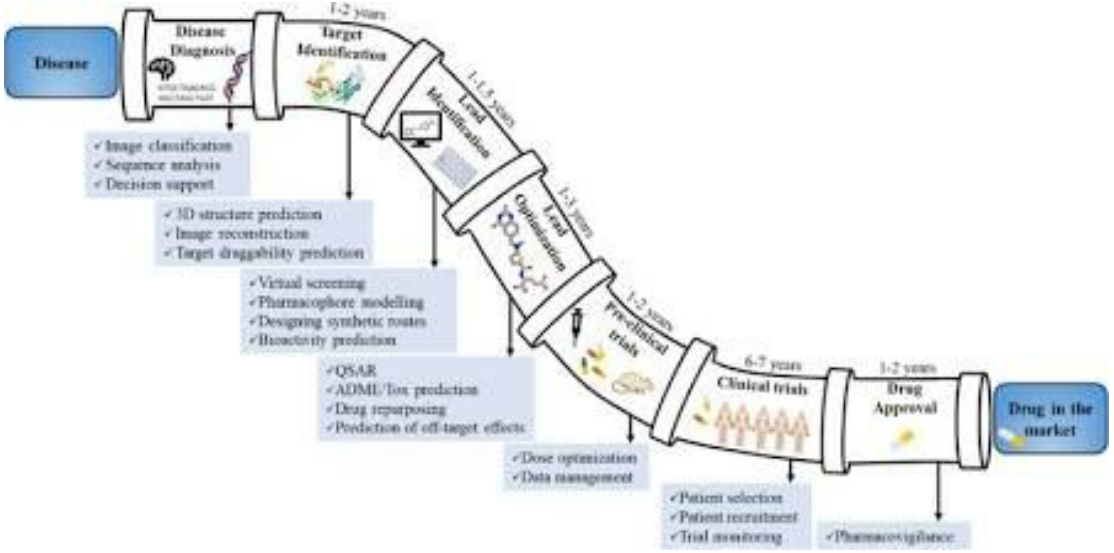


Drug potency

Drug type	**Potency**	
Small Molecule drugs	10 - 100 nM (nanomolar)	Cheap to manufacture, orally taken, long shelf life
Proteins (e.g. monoclonal Antibodies (mAbs))	0.1 - 10 nM	Expensive, intravenous, needs cooling and is perishable
Extraordinarily potent examples		
LSD (small molecule)	0.1 nM	Psychoactive drug
Botulinum Neurotoxin (protein)	100 fM - 1 pM (femto- to picomolar)	Most potent (and lethal) substance known

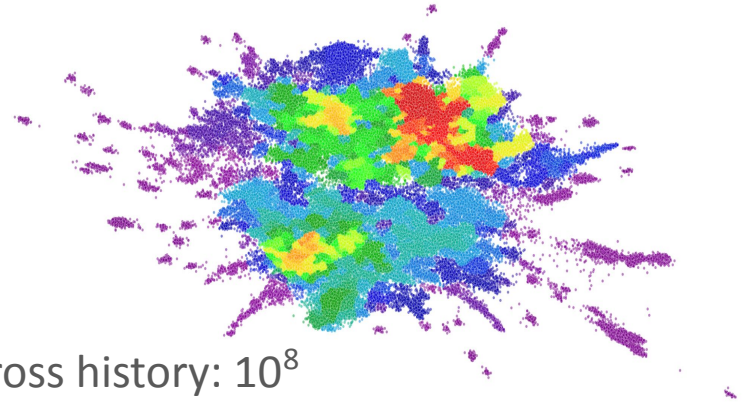
Our drugs are not as potent as nature's inventions

Drug Discovery



- **15 years & $15 billion**
- Target ID
- Hit Identification
- Lead Optimization (1–3 years)
- Preclinical Testing (1–2 years)
- Clinical Trials (6–7 years)
    a. Phase I – Safety, tolerability (healthy volunteers)
    b. Phase II – Efficacy and optimal dosing (100s patients)
    c. Phase III – Large-scale, randomized controlled trials (1000s patients)
- Regulatory Approval & Market Launch (1–2 years)

The Core Problem



Chemical space: ~10^{60} possible small molecules

Total compounds synthesized by all of humanity across history: 10^{8}

Total water molecules on earth: 10^{46}

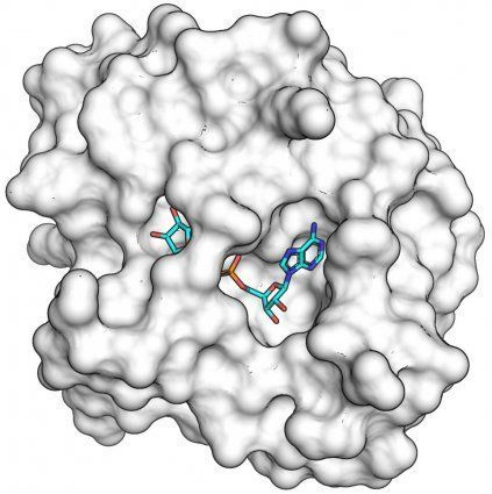
Find a small molecule with high binding affinity

Approaches:

1. High Throughput Screening (HTS) → Expensive, only on existing compound libraries
2. Molecular Dynamics (MD) → Takes forever on a supercomputer
3. AI → Getting better,...

AtomNet

- Proteins are large molecules, structure is important
- Small molecule fits in binding site and binds through complex bonds
- Question: Does it **bind**, or just merely *fit* there?



The SARS-CoV-2 macro domain protein bound to small molecule fragments that could be the basis of novel antiviral drugs.

The Atomwise approach (AtomNet)

1. Place (Smina Autodock) a candidate molecule in the protein binding pocket
2. Make a 3D voxelized image of the protein & molecule atoms
3. Feet examples in a 3D Convolutional Neural Network (CNN), have it predict binding affinity
4. 5000 proteins with on average 2000 compounds per protein, total 10 million training points

AtomNet – Model Performance

- AtomNet was a 99.9% accurate model[1]

> *“Our customers say the model it is not good enough, can you make it better?”*

1) I loosely speak of *accuracy*, in reality a metric like LogAUC is often used.

Challenge Accepted!



Feb 2018

The Random Test Split

- Split all data in 80% for training, 20% for testing – *Randomly*
- → 99.9% accuracy

What's possibly wrong?

Lead Optimization!

- After a Hit is identified, chemist make small modifications and try very similar molecules, in hope to optimize favorable properties

→ Many very similar molecules present in historical data

Better approach for train/test split?

Time Split

- All data measured before 1990 in the train set, everything after 1990 in the test set
- Model performance is now 90% accuracy

→ But many proteins are very alike as well, and have been studied for decades, crossing over the time split

Any ideas for further improvement?

Target Split

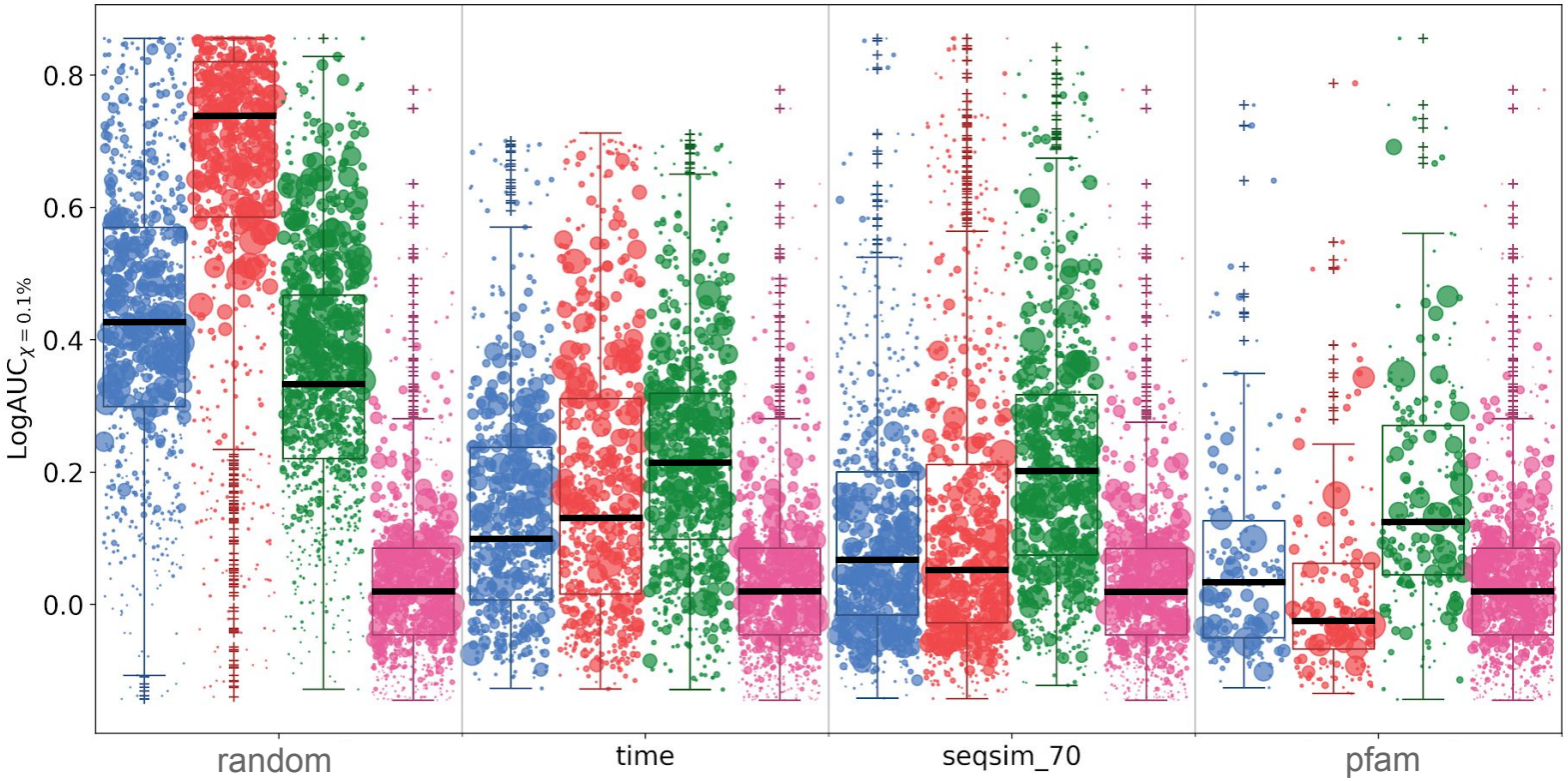
- 80% of the targets with their compounds in the train set, 20% in test
- Most proteins that have been researched are coming from a small class of proteins: Kinases, most proteins of the human genome are not studied at all. Many kinase inhibitors, inhibit many other kinases as well
- Improved target split: *Sequence similarity*, ensure similar targets are together in either the train or the test split
- Protein *PFAM clusters*, many small protein clusters in the train set, a few highly dissimilar clusters in test

→ Now the paper accuracy matched the customer experienced ‘accuracy’

Data Improvements

- **Publishing bias**: Most data is *positive* data, compound that didn't work aren't being published
- Adding the **Decoys**: take random compounds and fit them in random protein targets (assume they're non-binding (negative))
- Be careful with *promiscuous compounds*
- 3D Augmentations: rotate the protein-molecule pair on two axis

→ Our Computer-Aided Drug Design engineers noted: *Molecules with high rotational symmetry were rising to the top*



- Each bubble is a target (protein)
- Size of the bubble reflects the number of data points for that target
- LogAUC: AUC of the ROC curve with logarithmic X-axis

- Random Forest (RF) w pharmacophores
- RF w molecule-hashes (ECFP4)
- AtomNet
- Physics based potential model (SMINA)

Model Improvements

- Switched to a Graph Neural Network (GNN): Nodes are Atoms, Edges are distances to neighbouring atoms, within a certain *radius of influence*
- Introduced rotation invariant GNN, using spherical harmonics
- Conceptually much better:
  - No voxelization, but real atom positions
  - No rotation augmentation, but build-in symmetry
  - Atom - atom interactions explicitly encoded using edges, rather than the finite size of a convolution template

→ In practise the model didn’t perform much better than it’s CNN counterpart



Silicon Valley Economics

Silicon Valley Economics — Atomwise Case

$219M invested (2018–2020) • Demanding 10x VC returns by 2028

Capital Raised

$219M

Atomwise (2018–2020)

High-risk venture capital funds deployed during the peak AI drug discovery boom.

Target Return

$2.0B

Required Valuation by 2028

10x VC expectation requires ~$200M in annual income or ~$2B in annual revenue.

Investor Thesis

10x Bet

High-Risk Dynamics

What assumptions did investors rely on to justify scaling to a multi-billion dollar outcome?

The Core Strategic Question

How does this economic model hold up — and what is the underlying value of the AI platform really worth?

The Value — Drug Discovery

If we can generate a drug candidate at the click of a button, what is it actually worth?

Testing Pipeline

15 Years

Remaining Development

Still required to complete all clinical studies and regulatory approvals post-discovery.

Exclusivity Window

10–20 Yrs

Patent Expiration Lifespan

Commercial window ticks down rapidly while clinical testing consumes the majority of patent time.

Uncertainty

High Risk

Downstream Attrition

A 'hit' molecule still faces severe clinical failure rates before reaching market.

The Core Strategic Question

Would a pharmaceutical company pay top dollar for the discovery of a 'hit', knowing there is still 15 years of risk ahead?

The Value — Cost Reduction

What is the traditional alternative, and how does AI enrichment transform screening economics?

Traditional HTS

200k – 2M

Compounds tested per assay
Cost Baseline: **~$1 / compound**
Pre-set compound library

Typical Hit Rate **~4%**

AI Model Value 2X ENRICHMENT

4% → 8%

Enriched hit rate per screening set
50% Cost Savings
Need to screen only **half as many compounds** for the same result.
Direct bottom-line efficiency gain in assay discovery

Total Addressable Market

$2.5B TAM

~5 Billion
Global annual raw HTS well-plate readouts (today)

Practical Reality & Implementation Nuances

1. **Synthesis Limits:** Not every compound can easily be made or ordered off-the-shelf.
2. **Operational Simplicity:** Often easier to screen entire library than to subset.
3. **Diversity Focus:** Better to screen highly *dissimilar* molecules to obtain the broadest coverage of chemical space.

The Value — Searching Chemical Space

How super-human AI models navigate vast molecular landscapes to uncover hyper-potent drugs.

AI EXPLORATION ENGINE

AI Model Becomes Super-Human

- Explore vastly larger parts of chemical space
- Discover true needle in the haystack molecules
- **Super potent drugs**

FOR PERSPECTIVE

10 pM Small Molecule Drug

A 10 pM small molecule drug would have full pharmacological effect at a dosing below **Threshold of Toxicological Concern (TTC)**.

FDA and EMA regulatory framework

EFFICIENCY

Reduce Time to Market

By skipping some testing, not needed for ultra low dosing

SAFETY

Reduce Side Effects

Highly potent & targeted drugs are less likely to bind to other proteins

NOVELTY

Drug the Undruggable

Finding truly novel compounds to solve previously untreatable disease

Values — The Need for a Moat

Why core innovation capabilities require defensible competitive moats to create investor value.

CORE VALUE DRIVERS

Key Capabilities

- Drug Discovery
- HTS Cost Reduction
- Alien Drugs (Exploring Chemical Space)

THE PROTECTION TRAP

Standard IP Decays

- Employees move
- Secrets leak
- Patents expire (and get work-arounds)

None of these Values is worth anything to an investor without a Moat

Real venture value requires feedback loops and network effects that compound faster than traditional IP erodes.

The Network Effects — The Data Flywheel

How ML/AI in drug discovery constructs exponential competitive moats.

01 | FIRST MOVER

Early Mover

ML/AI for Drug Discovery

02 | DATA PLAY

Data Play

Big pharma gives their proprietary historical HTS data, in exchange for predictions and lower HTS costs.

03 | FLYWHEEL

Flywheel

Building bigger models, with more data, leading to better predictions.

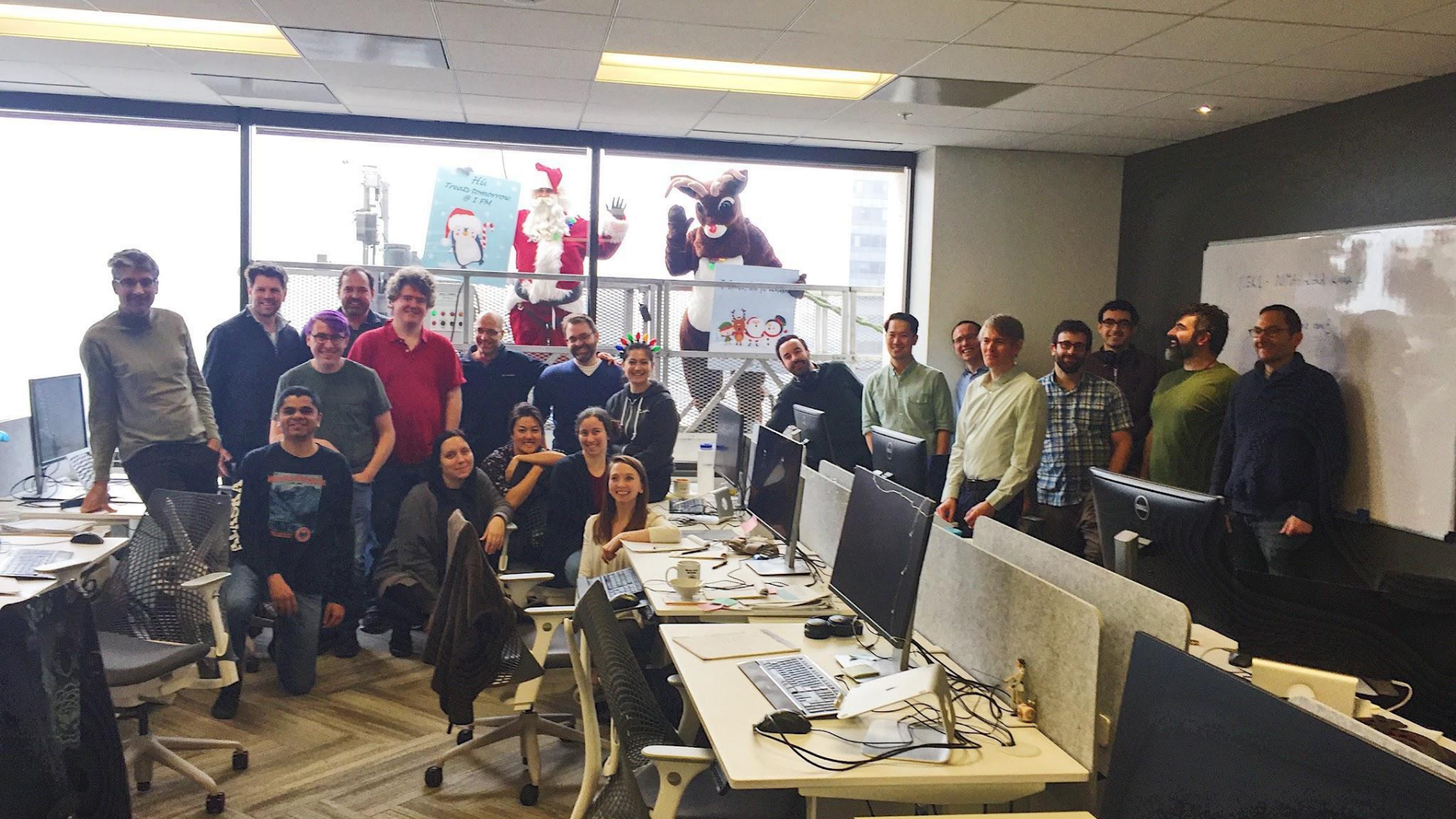
04 | REPEAT

Continuous Cycle

Repeat the cycle to compound data advantage and expand market lead.

True ‘Silicon Valley’ value comes from creating a moat, which increases exponentially with the value itself.

Proprietary data loops yield non-linear defensibility that compounds faster than traditional IP decays.


Hi
Treats Tomorrow
@ 1 PM

Modern Advancements in AI4DD

Alpha Fold – (July 2021) Break through in protein structure prediction, but heavily relies on Multiple Sequence Alignment (MSA)

Co-folding – Alpha-fold + drug, see if and how it will fold

Cryo-EM – Emerging technique to better and faster measure protein structures

Diffusion models – Instead of searching through chemical space, trying one molecule at a time, build up a molecule in-place (inside the binding pocket)

Machine-Learning Force Fields (MLFFs) – Speed up MD by training AI on DFT calculations, quantum force fields at the speed of classical ones